

reWatt

User Guide



FIRE ALARM

AI ENERGY OPTIMISATION

SECURITY CAMERA

Android 6.0+ any old phone • **0** rewiring • **100%** on-device AI

WELCOME

Your old phone, now a smart room sensor.

reWatt turns an old Android phone into three helpers in one.

Clip the reWatt module onto the phone. Mount it on the wall. That's it.

Android 6.0+
any old phone

0 rewiring
magnetic or adhesive

100% on-device AI
nothing uploaded

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BEFORE YOU START

What you need

reWatt module
With its adapter and splitter.

An old Android phone
Android 6.0 or newer. The camera must work.

Phone charger
A USB-C charger and cable.

Wi-Fi
For setup and MQTT.

A wall near a power socket
Any wall works with the adhesive mount. A metal surface works with the magnet.



The module clips onto the back of any old phone. No rewiring.

STEP 1

Prepare your old phone

1. **Charge it.**
   Charge the phone fully before you start.
2. **Clear it.**
   Back up your old photos. Then reset the phone or delete them.
3. **Connect to Wi-Fi.**
   Use the Wi-Fi of the room it will watch.
4. **Update it.**
   Install any Android and app updates.
5. **Allow the install.**
   Settings → Apps → Special access → *Install unknown apps*. Allow your browser or file app.

STEP 2

Install the app

1. **Download.**
   Get the reWatt app file (*.apk*) from the link that came with your kit.
2. **Install.**
   Tap the file, then tap *Install*.
3. **Open.**
   Tap *Open*. The Quick Setup Guide starts by itself.

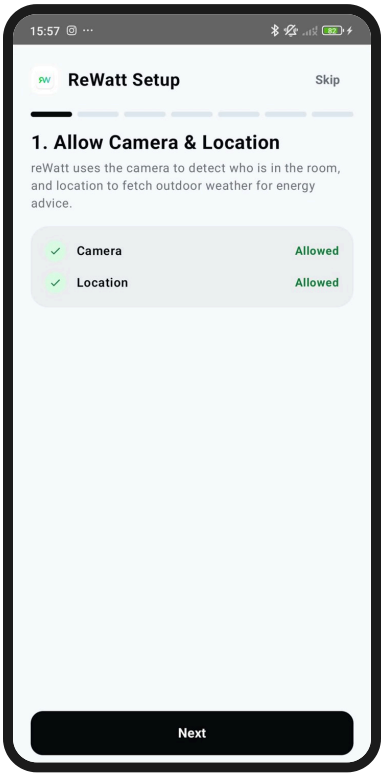
> **TIP**
> Android may warn about unknown apps. This is normal for apps not from the Play Store.

STEP 3 · QUICK SETUP GUIDE

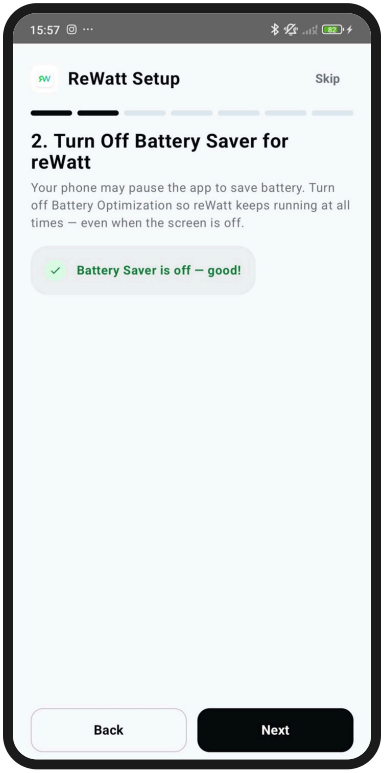
Set up the app

The guide has 7 short pages. Tap **Next** to move on.

1 Allow Camera & Location
The camera sees who is in the room. Location gets the outdoor weather. Tap *Allow* on both.



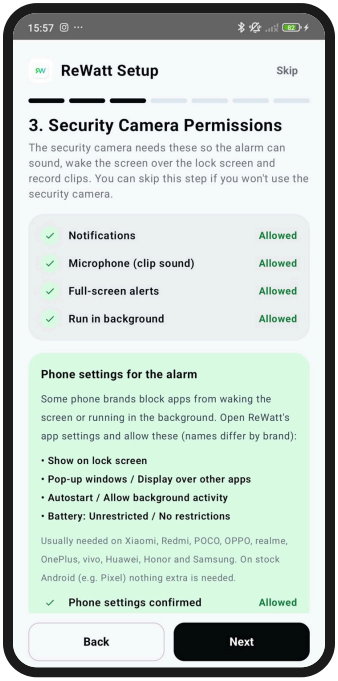
2 Turn off Battery Saver
Otherwise the phone may pause reWatt. Set it to *Off* for reWatt.



STEP 3 · QUICK SETUP GUIDE

3 **Security permissions**

Allow each item until all say *Allowed*. The alarm needs them to ring and wake the screen.

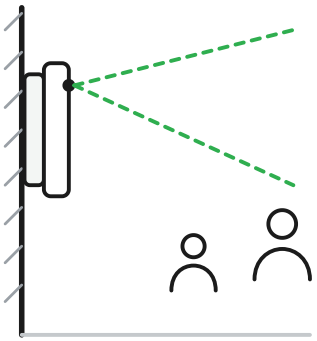


Some brands (Xiaomi, OPPO, vivo, Samsung…) need extra settings. The page lists them.

4 **Mount the phone**

Attach it to a wall or metal surface. Point the camera at the centre of the room.

- Shoulder height or higher.
- Camera faces the seating area.
- Keep the lens clear. No tape or cover.
- Keep the charger plugged in.

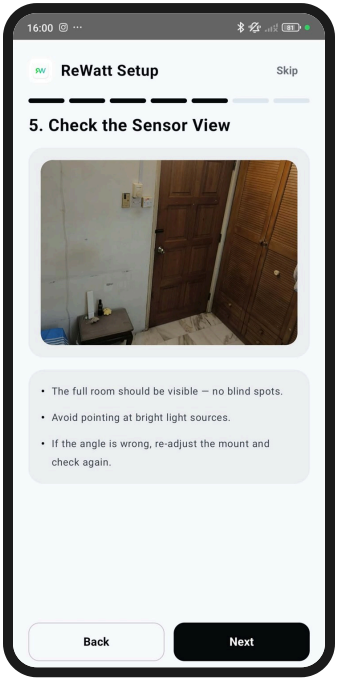


Mount at shoulder height, camera facing the room.

STEP 3 · QUICK SETUP GUIDE

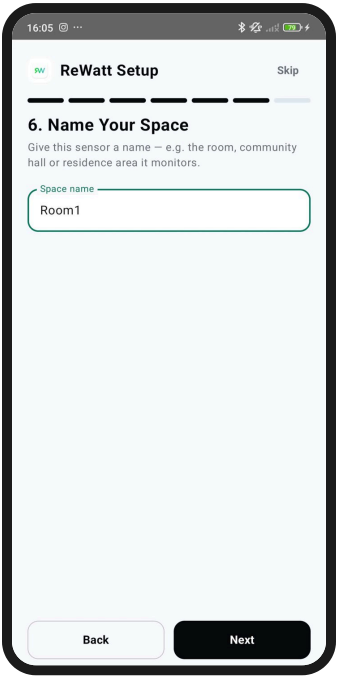
5 **Check the view**

The whole room should be visible. Avoid bright lights. Move the mount if needed.



6 **Name your space**

Type a name, e.g. *Room1*, *Living room* or *Community hall*.



7 **You're all set!**

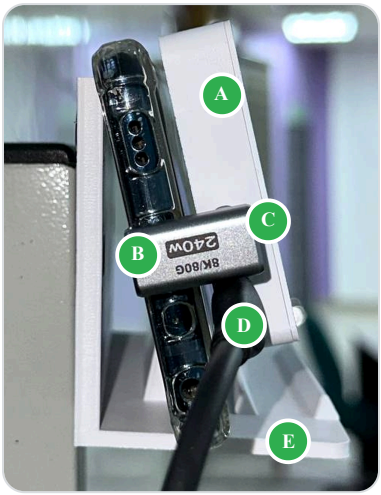
Tap *Done*. reWatt now watches the room for you.

OPEN IT AGAIN

Settings → *Quick Setup Guide*.

STEP 4

Connect the module



A **reWatt module**
Sits behind the phone.

B **Phone's charging port**
The 180° adapter plugs in here.

C **Module port**
The adapter's other end goes into the module.

D **Charging port**
Plug your phone charger in here.

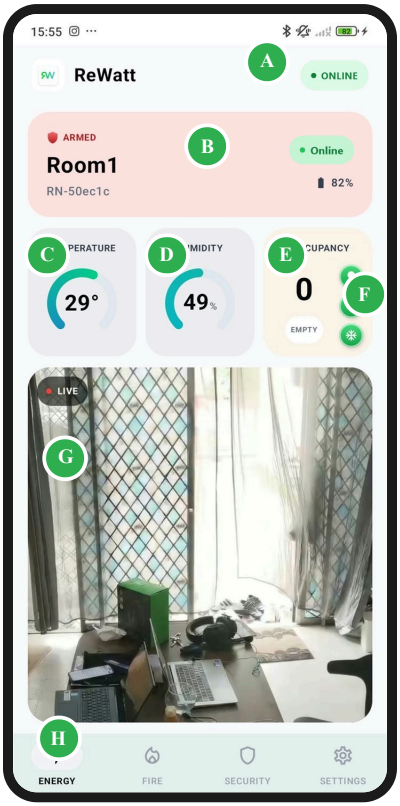
E **Wall mount**
Holds the phone and module.

1. **Clip it on.**
   Place the module on the back of the phone. Use the magnet or adhesive pad.
2. **Plug into the phone.**
   Push the *Type-C 180° adapter* into the phone's charging port (B) and the module port (C).
3. **Plug in power.**
   Connect your phone charger to the charging port (D).
4. **Allow it.**
   If Android asks to open reWatt for the USB device, tap *OK*. Tick *Always* if shown.
5. **Check it.**
   The badge at the top right changes from *OFFLINE* to **ONLINE**.

> ⚠ **IMPORTANT**
> Keep the charger connected at all times. The phone runs 24/7.

USING REWATT

The ENERGY screen



A **ONLINE / OFFLINE**
Is the module connected?

B **Room card**
Room name, security status and battery.

C **Temperature**
Room temperature in °C.

D **Humidity**
Moisture in the air, in %.

E **Occupancy**
People in the room. *EMPTY* at 0.

F **Lights · Fan · AC**
Green = on. Red = off.

G **Live view**
What the camera sees right now.

H **Menu**
ENERGY · FIRE · SECURITY · SETTINGS

TIP
Turn the phone sideways for a bigger live view.

USING REWATT

Your readings explained

READING	WHAT IT MEANS
Temperature	How warm the room is, in °C. Measured by the SHT31.
Humidity	How damp the air is, in %. About 40–60% feels comfortable.
Occupancy	How many people the camera counts. The radar wakes the camera when someone is there.
Lights · Fan · AC	Shows if each appliance is on (green) or off (red).
VOC Index	Gases from overheating wires or a gas leak. Normal is about **100**.
NOx Index	Gases from something burning. Normal is about **1**.

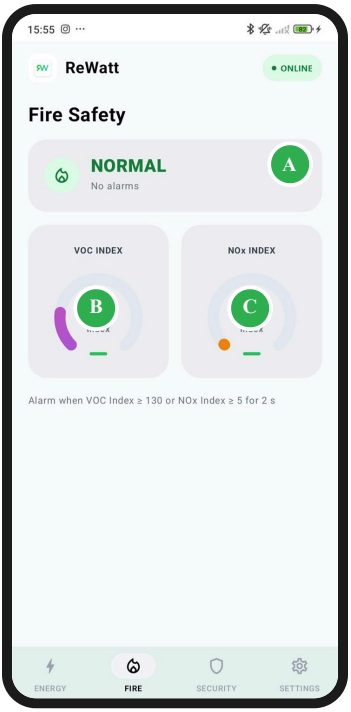
How reWatt saves energy

- People in the room → reWatt picks the right AC setting for the headcount.
- More people in the room → reWatt lowers the AC temperature to keep everyone cool.
- No one for **5 minutes** → the room's lights, fan and AC switch off.
- Empty room → camera and AI stay off to save power.

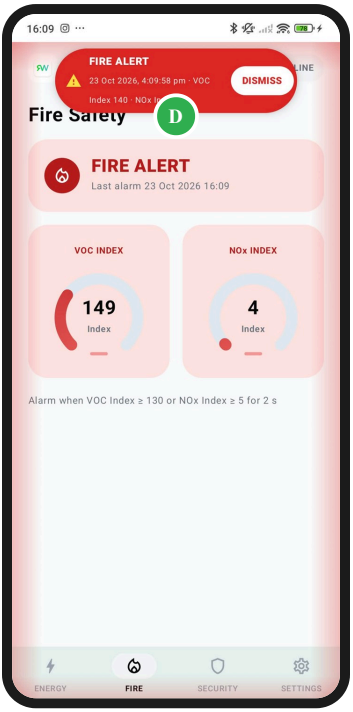
GOOD TO KNOW
You can also see these readings on another device. See *Connect to MQTT*, page 13.

USING REWATT

The FIRE screen



Normal



Fire alert

A **Status**
NORMAL or FIRE ALERT.

B **VOC Index**
Earliest warning. Normal ≈ 100.

C **NOx Index**
Something is burning. Normal ≈ 1.

D **Alert bar**
Time, readings and DISMISS.

When does the alarm ring?

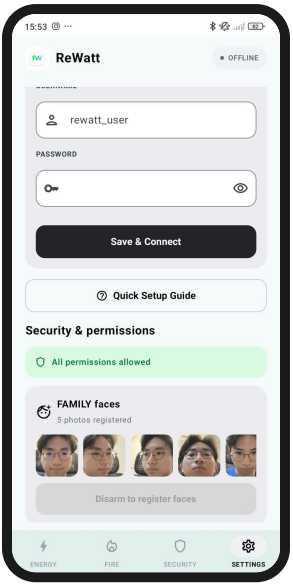
Normal air — VOC 130+ or NOx 5+ for 2 seconds → ALARM

IF THE ALARM RINGS
Get everyone out. Call 994 (Bomba). Only tap DISMISS when you are safe.

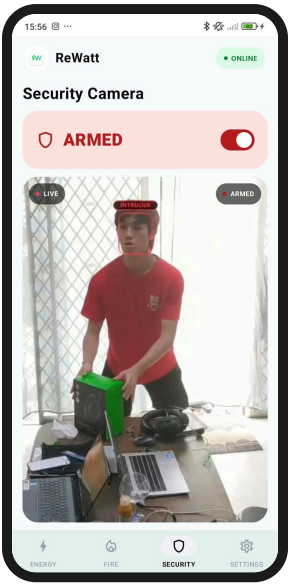
The fire alarm is always on, even when security is off. Test it in Settings → *Test fire alarm*.

USING REWATT

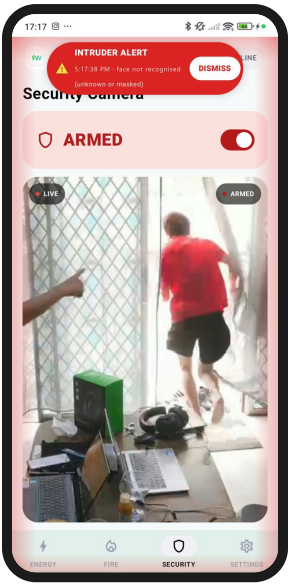
The SECURITY screen



Settings → FAMILY faces



Armed: a stranger is seen



Siren + intruder alert

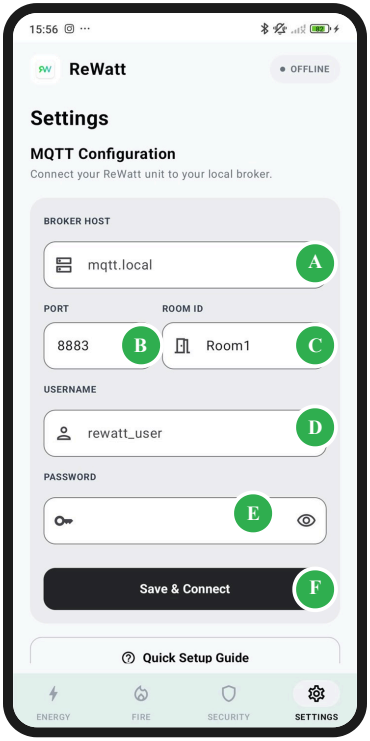
1. **Add family**
   Settings → *FAMILY faces*. Add a few photos of each person.
2. **Arm it**
   Open SECURITY. Turn on *ARMED* when you leave or at night.
3. **Family?**
   reWatt stays silent.
4. **Stranger?**
   It says *"You are being recorded"*. Then the siren, flashing light and alert start.
5. **Stop it**
   Tap DISMISS.

PRIVATE BY DESIGN
Faces are checked on the phone. Nothing is uploaded. Turn off ARMED to add or change faces. Past alerts are listed below the camera.

SETTINGS

Connect to MQTT

MQTT sends your readings and alerts to your own server.



A **Broker host**
Your MQTT server address.

B **Port**
Use **8883** (secure).

C **Room ID**
A short room name, e.g. *Room1*.

D **Username**
From your MQTT server.

E **Password**
Tap the eye to check it.

F **Save & Connect**
Tap to save and connect.

1. Open **SETTINGS** → *MQTT Configuration*.
2. Fill in A to E.
3. Tap *Save & Connect*.

Readings can then be viewed on another device.

HOW IT WORKS

One radar wakes everything. Three jobs, one phone.

A radar senses people, even when they sit still. The camera only wakes up when someone is there.



Energy optimisation

Presence
Someone is in the room

›

Headcount
Counts people

›

Temp & humidity
Checks the air

›

Right AC setting
Off at 0 people

No one for 5 minutes → the room's appliances switch off.



Security camera

Presence
Someone is in the room

›

Person seen
No face → it records

›

Face check
On the phone

›

Family?
✓ Stays silent
✗ Flashing light + alert



Fire alarm · always on

Gas sensing
24/7

›

VOC spike
Early warning

›

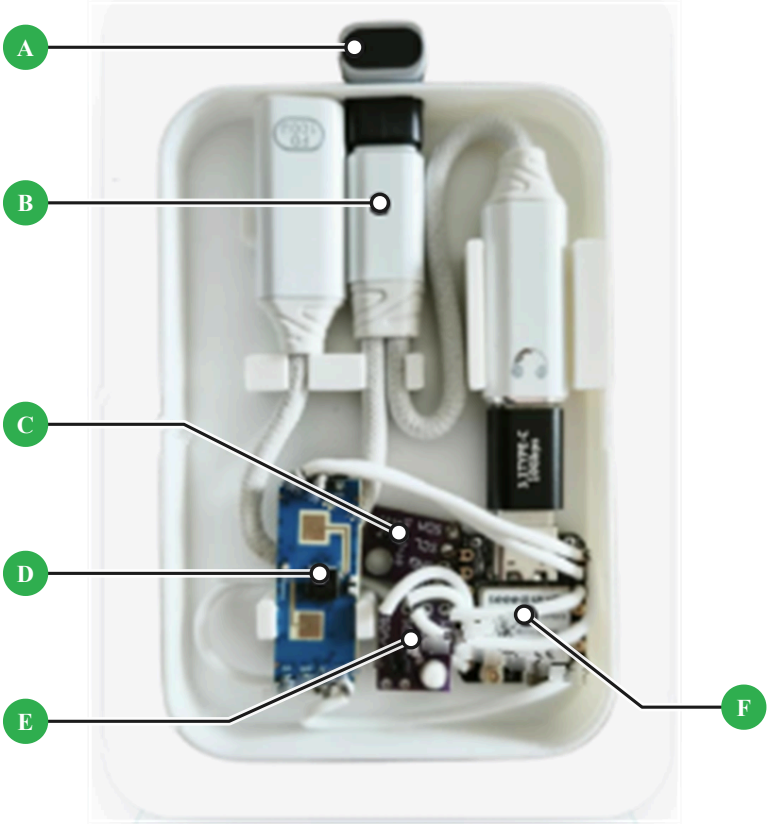
NOx rise
Something is burning

›

Loud siren
+ alert

WHAT'S INSIDE

Module components



A **Type-C 180° adapter**
Plugs the module into the phone.

B **Type-C 1-to-2 splitter**
Charge + USB. Charges the phone and talks to it.

C **SGP41 · gas sensor**
Senses VOC and NOx gases, 24/7.

D **LD2412 · mmWave radar**
Senses people, even sitting still.

E **SHT31 · climate sensor**
Measures temperature and humidity.

F **ESP32-C6 · controller**
All three sensors connect to it.

HELP

Quick fixes

PROBLEM	TRY THIS
OFFLINE	Check the adapter is pushed in. Unplug and plug it back in.
No live view	Allow the camera permission. Clean the lens.
App stops at night	Turn off Battery Saver for reWatt (setup page 2).
Alarm is silent	Allow all security permissions (setup page 3).
Wrong headcount	Move the phone so the whole room is in view.
Family member alerted	Add more photos of that person in FAMILY faces.

Care & safety

- Use indoors only. Keep it dry.
- Do not cover the module's vents or the camera.
- Use a good-quality charger and cable.
- Do not open the module or bend its parts.
- reWatt helps you stay safe. It does not replace a certified smoke detector.